

# US Patent & Trademark Office

## Patent Public Search | Text View

---

|                      |                    |
|----------------------|--------------------|
| United States Patent | 12409840           |
| Kind Code            | B2                 |
| Date of Patent       | September 09, 2025 |
| Inventor(s)          | Kumai; Yuichi      |

---

### Vehicle control apparatus

---

#### Abstract

A vehicle control apparatus executes a lane change control of providing a driver of an own vehicle with a proposal of changing lanes and automatically changes lanes when the driver approves of the proposal. When the driver performs an acceleration operation when the vehicle control apparatus provides the driver with a proposal of changing lanes to move the own vehicle to a next lane for overtaking a preceding vehicle, the vehicle control apparatus determines that the driver approves of the proposal. When the driver performs a deceleration operation when the vehicle control apparatus provides the driver with a proposal of changing lanes to return the own vehicle to an original lane after the own vehicle moves to the next lane and overtakes the preceding vehicle, the vehicle control apparatus determines that the driver approves of the proposal.

---

|                   |                                              |
|-------------------|----------------------------------------------|
| <b>Inventors:</b> | Kumai; Yuichi (Numazu, JP)                   |
| <b>Applicant:</b> | TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota, JP) |
| <b>Family ID:</b> | 93381082                                     |
| <b>Assignee:</b>  | TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota, JP) |
| <b>Appl. No.:</b> | 18/641995                                    |
| <b>Filed:</b>     | April 22, 2024                               |

#### Prior Publication Data

|                            |                         |
|----------------------------|-------------------------|
| <b>Document Identifier</b> | <b>Publication Date</b> |
| US 20240375660 A1          | Nov. 14, 2024           |

#### Foreign Application Priority Data

|    |             |               |
|----|-------------|---------------|
| JP | 2023-077242 | May. 09, 2023 |
|----|-------------|---------------|

---

#### Publication Classification

**Int. Cl.: B60W30/18** (20120101); **B60W50/10** (20120101)

**U.S. Cl.:**

**CPC** **B60W30/18163** (20130101); **B60W50/10** (20130101); B60W2554/802 (20200201); B60W2720/24 (20130101)

## Field of Classification Search

**CPC:** B60W (30/18163); B60W (50/10); B60W (2554/802); B60W (2720/24)

---

## References Cited

### U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name    | U.S. Cl. | CPC          |
|--------------|-------------|------------------|----------|--------------|
| 10793147     | 12/2019     | Kaminade et al.  | N/A      | N/A          |
| 10870349     | 12/2019     | Uejima           | N/A      | N/A          |
| 10953883     | 12/2020     | Sakaguchi        | N/A      | N/A          |
| 10981569     | 12/2020     | Hashimoto et al. | N/A      | N/A          |
| 11014563     | 12/2020     | Hashimoto et al. | N/A      | N/A          |
| 11072334     | 12/2020     | Aoki et al.      | N/A      | N/A          |
| 11267473     | 12/2021     | Takahashi        | N/A      | N/A          |
| 11279360     | 12/2021     | Fukuda et al.    | N/A      | N/A          |
| 2019/0107835 | 12/2018     | Hashimoto        | N/A      | N/A          |
| 2019/0202471 | 12/2018     | Hashimoto        | N/A      | N/A          |
| 2020/0070889 | 12/2019     | Fukuda et al.    | N/A      | N/A          |
| 2020/0247400 | 12/2019     | Fujii            | N/A      | N/A          |
| 2021/0284164 | 12/2020     | Hirosawa         | N/A      | B60W 60/0055 |
| 2022/0234586 | 12/2021     | Oniwa et al.     | N/A      | N/A          |
| 2024/0132068 | 12/2023     | Hempel           | N/A      | B60W 30/146  |

### FOREIGN PATENT DOCUMENTS

| Patent No.  | Application Date | Country | CPC |
|-------------|------------------|---------|-----|
| 2019-069711 | 12/2018          | JP      | N/A |
| 2019-119269 | 12/2018          | JP      | N/A |
| 2022-113048 | 12/2021          | JP      | N/A |
| 2022-128973 | 12/2021          | JP      | N/A |

---

*Primary Examiner:* Mo; Xiao En

*Attorney, Agent or Firm:* Oliff PLC

---

## Background/Summary

### CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to Japanese patent application No. JP 2023-077242 filed on May 9, 2023, the content of which is hereby incorporated by reference in its entirety.

### BACKGROUND

Field

(2) The present invention relates to a vehicle control apparatus.

#### Description of the Related Art

(3) There is known a vehicle control apparatus which is configured to propose changing lanes to a driver of an own vehicle, determine that the driver approves of a proposal of changing lanes when the driver operates a predetermined button, and automatically change lanes on which the own vehicle is moving (for example, refer to JP 2022-113048 A).

(4) In the vehicle control apparatus described above, the driver needs to operate the predetermined button for approving of the proposal of changing lanes. An operation applied to the predetermined button is not an operation necessary for the driver to travel the own vehicle such as an operation to accelerate or decelerate the own vehicle. Therefore, the driver may feel bothersome.

#### SUMMARY

(5) An object of the present invention is to provide a vehicle control apparatus which enable the driver approve of a proposal of changing lanes for the own vehicle by a simple operation.

(6) A vehicle control apparatus according to the present invention comprises an electronic control unit which executes a lane change control of providing a driver of an own vehicle with a proposal of changing lanes and automatically change lanes when the driver approves of the proposal. The electronic control unit is configured to, when the driver performs an acceleration operation when the electronic control unit provides the driver with a proposal of changing lanes to move the own vehicle to a next lane for overtaking a preceding vehicle, determine that the driver approves of the proposal of changing lanes to move the own vehicle to the next lane. Further, the electronic control unit is configured to, when the driver performs a deceleration operation when the electronic control unit provides the driver with a proposal of changing lanes to return the own vehicle to an original lane after the own vehicle moves to the next lane and overtakes the preceding vehicle, determine that the driver approves of the proposal of changing lanes to return the own vehicle to the original lane.

(7) With the vehicle control apparatus according to the present invention, when the driver is provided with the proposal of changing lanes to move the own vehicle to the next lane, the driver can approve of the proposal by performing the acceleration operation. In general, when the driver makes a lane change to move the own vehicle to the next lane by their driving operation, the driver performs the acceleration operation. Therefore, the acceleration operation performed by the driver is a simple operation as an operation for approving of the proposal of changing lanes to move the own vehicle to the next lane. In addition, with the vehicle control apparatus according to the present invention, when the driver is provided with the proposal of changing lanes to return the own vehicle to the original lane, the driver can approve of the proposal by performing the deceleration operation. In general, when the driver makes a lane change to return the own vehicle to the original lane by their driving operation, the driver performs the deceleration operation. Therefore, the deceleration operation performed by the driver is a simple operation as an operation for approving of the proposal of changing lanes to return the own vehicle to the original lane. Thus, the driver can approve of the proposal of changing lanes by performing a simple operation.

(8) In the vehicle control apparatus according to an aspect of the present invention, the electronic control unit may be further configured to, when the driver performs the deceleration operation when the electronic control unit provides the driver with the proposal of changing lanes to move the own vehicle to the next lane, determine that the driver rejects the proposal of changing lanes to move the own vehicle to the next lane. In this aspect, the electronic control unit may be further configured to, when the driver performs the acceleration operation when the electronic control unit provides the driver with the proposal of changing lanes to return the own vehicle to the original lane, determine that the driver rejects the proposal of changing lanes to return the own vehicle to the original lane.

(9) As described above, In general, when the driver makes a lane change to move the own vehicle

to the next lane by their driving operation, the driver performs the acceleration operation. Therefore, the deceleration operation performed by the driver is a simple operation as an operation for rejecting the proposal of changing lanes to move the own vehicle to the next lane. In addition, in general, when the driver makes a lane change to return the own vehicle to the original lane by their driving operation, the driver performs the deceleration operation. Therefore, the acceleration operation performed by the driver is a simple operation as an operation for rejecting the proposal of changing lanes to return the own vehicle to the original lane. Thus, with the vehicle control apparatus according to this aspect of the present invention, the driver can reject the proposal of changing lanes by performing a simple operation.

(10) In the vehicle control apparatus according to another aspect of the present invention, the electronic control unit may be further configured to provide the driver with the proposal of changing lanes to move the own vehicle to the next lane when a distance between the own vehicle and the preceding vehicle becomes equal to or smaller than a predetermined distance, and a travelling speed of the own vehicle is greater than a travelling speed of the preceding vehicle.

(11) The vehicle control apparatus according to this aspect of the present invention can determine whether to provide the proposal of changing lanes to move the own vehicle to the next lane based on the distance between the own vehicle and the preceding vehicle, and the travelling speeds of the own vehicle and the preceding vehicle.

(12) In the vehicle control apparatus according to further another aspect of the present invention, the electronic control unit may be further configured to provide the driver with the proposal of changing lanes to return the own vehicle to the original lane when a space where the own vehicle can enter exists ahead of the preceding vehicle after the vehicle moves to the next lane and overtakes the preceding vehicle.

(13) The vehicle control apparatus according to this aspect of the present invention can determine whether to provide the proposal of changing lanes to return the own vehicle to the original lane by determining whether a space where the own vehicle can enter exists ahead of the preceding vehicle.

(14) Elements of the invention are not limited to elements of embodiments and modified examples of the invention described with reference to the drawings. The other objects, features and accompanied advantages of the invention can be easily understood from the embodiments and the modified examples of the invention.

---

## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a view which shows a vehicle control apparatus according to an embodiment of the present invention.

(2) FIG. 2 is a view which shows a flowchart of a routine executed by the vehicle control apparatus according to the embodiment of the present invention.

(3) FIG. 3 is a view which shows a scene that an overtaking lane change proposal is provided.

(4) FIG. 4 is a view which shows a scene that a returning lane change proposal is provided.

### DETAILED DESCRIPTION

(5) Below, a vehicle control apparatus according to an embodiment of the present invention will be described with reference to the drawings. As shown in FIG. 1, the vehicle control apparatus **10** according to the present invention is mounted on an own vehicle **100**. The vehicle control apparatus **10** may be mounted on a vehicle that can be driven remotely.

(6) Further, the present invention can be applied not only to a manual driving vehicle which is driven by manual driving in which a driver performs a driving operation on the own vehicle **100** by himself/herself, but also to an automatic driving vehicle which is driven by automatic driving in which a control device such as an ECU automatically drives the own vehicle **100** without the driver

performing the driving operation by himself/herself.

(7) As shown in FIG. 1, the vehicle control apparatus **10** includes an ECU (an electronic control device) **90** as a control device. The ECU **90** includes a microcomputer as a main component. The microcomputer includes a storage medium such as a CPU, a ROM, a RAM and a non-volatile memory, an interface, and the like. The CPU is configured or programmed to realize various functions by executing instructions, programs, or routines stored in the storage medium. In particular, in the present embodiment, the vehicle control apparatus **10** stores, in the storage medium, programs for realizing various kinds of controls executed by the vehicle control apparatus **10**.

(8) It should be noted that the vehicle control apparatus **10** may be configured to be able to update the programs stored in the storage medium by wireless communication (for example, Internet communication) with external devices.

(9) As shown in FIG. 1, the own vehicle **100** is equipped with a driving apparatus **21**, a braking apparatus **22**, and a steering apparatus **23**. The driving apparatus **21**, the braking apparatus **22**, and the steering apparatus **23** are electrically connected to the ECU **90**. The vehicle control apparatus **10** controls a driving force applied to the own vehicle **100** by controlling operations of the driving apparatus **21**. Further, the vehicle control apparatus **10** controls a braking force applied to the own vehicle **100** by controlling operations of the braking apparatus **22**. Further, the vehicle control apparatus **10** controls a steering force applied to the own vehicle **100** by controlling operations of the steering apparatus **23**.

(10) Further, the own vehicle **100** is equipped with a proposal device **30**, an accelerator pedal **41** as an acceleration operation device, a brake pedal **42** as a deceleration operation device, a set vehicle travelling speed increase button **43** as a set vehicle travelling speed increase operation device, a set vehicle travelling speed decrease button **44** as a set vehicle travelling speed decrease operation device, a vehicle travelling speed detection device **45**, and a surrounding information detection apparatus **50**.

(11) The proposal device **30** is a device which provides a proposal of changing lanes of the own vehicle **100**, which will be described later, to a driver of the own vehicle **100**. In the present embodiment, the proposal device **30** includes a display **31** and a speaker **32**. The display **31** as a displaying device and the speaker **32** as an audio device are electrically connected to the ECU **90**, respectively. The vehicle control apparatus **10** displays a lane change proposal image, which will be described later, on the display **31**, and outputs lane change proposal sound, which will be described later, from the speaker **32**.

(12) The accelerator pedal **41** is a device operated by the driver of the own vehicle **100** to accelerate the own vehicle **100**. The accelerator pedal **41** is electrically connected to the ECU **90**. The vehicle control apparatus **10** acquires an operation amount of the accelerator pedal **41** and controls the driving force applied to the own vehicle **100** in accordance with the operation amount.

(13) The brake pedal **42** is a device operated by the driver to decelerate the own vehicle **100**. The brake pedal **42** is electrically connected to the ECU **90**. The vehicle control apparatus **10** acquires an operation amount of the brake pedal **42** and controls the braking force applied to the own vehicle **100** in accordance with the operation amount.

(14) The set vehicle travelling speed increase button **43** is a device operated by the driver to increase a set vehicle travelling speed  $V_{set}$  in a constant speed travel control. The constant speed travel control is a control of automatically maintaining an own vehicle travelling speed  $V_{ego}$  at the set vehicle travelling speed  $V_{set}$ . The set vehicle travelling speed increase button **43** is electrically connected to the ECU **90**. The vehicle control apparatus **10** increases the set vehicle travelling speed  $V_{set}$  when the set vehicle travelling speed increase button **43** is operated.

(15) The set vehicle travelling speed decrease button **44** is a device operated by the driver to decrease the set vehicle travelling speed  $V_{set}$  in the constant speed travel control. The set vehicle travelling speed decrease button **44** is electrically connected to the ECU **90**. The vehicle control

apparatus **10** decreases the set vehicle travelling speed  $V_{set}$  when the set vehicle travelling speed decrease button **44** is operated.

(16) The vehicle travelling speed detection device **45** is electrically connected to the ECU **90**. The vehicle control apparatus **10** acquires a traveling speed of the own vehicle **100** as the own vehicle travelling speed  $V_{ego}$  by the vehicle travelling speed detection device **45**. The vehicle travelling speed detection device **45** includes, for example, wheel rotation speed sensors.

(17) The surrounding information detection apparatus **50** is an apparatus which detects information on situations around the own vehicle **100**. In the present embodiment, the surrounding information detection apparatus **50** includes image sensors **51** and electromagnetic wave sensors **52**.

(18) The image sensor **51** is a sensor which captures and acquires images of views around the own vehicle **100**. The image sensor **51** is, for example, a camera. The image sensors **51** are electrically connected to the ECU **90**, respectively. The vehicle control apparatus **10** acquires images (or surrounding images) around the own vehicle **100** as surrounding detection information IF by the image sensors **51**.

(19) The electromagnetic wave sensor **52** is a sensor which detects objects existing around the own vehicle **100**. The electromagnetic wave sensor **52** is, for example, a radio wave sensor such as a radar sensor such as a millimeter wave radar, a sound wave sensor such as an ultrasonic sensor such as a clearance sonar, or an optical sensor such as a laser radar such as a LIDAR. The electromagnetic wave sensors **52** are electrically connected to the ECU **90**, respectively. The vehicle control apparatus **10** acquires, by the electromagnetic wave sensors **52**, information (or object information) related to the objects around the own vehicle **100** as the surrounding detection information IF.

(20) <Operations of Vehicle Control Apparatus>

(21) Next, operations of the vehicle control apparatus **10** will be described. The vehicle control apparatus **10** executes a lane change control of automatically changing lanes of the own vehicle **100** when a predetermined condition becomes satisfied by executing a routine shown in FIG. 2 at a predetermined calculation cycle.

(22) At a predetermined timing, the vehicle control apparatus **10** starts a process from a step S200 of the routine shown in FIG. 2 and proceeds with the process to a step S205 to determine whether or not an overtaking lane change condition C1 is satisfied.

(23) As shown in FIG. 3, the overtaking lane change condition C1 is a condition that the driver is presumed to desire to perform an overtaking lane change control to be described later. More specifically, the overtaking lane change condition C1 is, for example, a condition that an inter-vehicle distance  $D$ , i.e., a distance between the own vehicle **100** and a preceding vehicle **200** is equal to or smaller than a predetermined distance (or a predetermined inter-vehicle distance  $D_{th}$ ) and the own vehicle travelling speed  $V_{ego}$  is greater than a traveling speed  $V_{pre}$  of the preceding vehicle **200** by a predetermined speed  $\Delta V$  or more ( $V_{ego} \geq V_{pre} + \Delta V$ ). The vehicle control apparatus **10** acquires the inter-vehicle distance  $D$  and the traveling speed  $V_{pre}$  of the preceding vehicle **200** based on the surrounding detection information IF. The overtaking lane change condition C1 may include a condition that a lane keeping control is being executed. The lane keeping control is a control of assisting a steering of the own vehicle **100** such that the own vehicle **100** travels while maintaining a center position of a lane. Further, the predetermined speed  $\Delta V$  is set to a value greater than zero.

(24) When the vehicle control apparatus **10** determines “Yes” at the step S205, the vehicle control apparatus **10** proceeds with the process to a step S210 to execute a proposal process of an overtaking lane change. The proposal process of the overtaking lane change is a process of providing the driver with an overtaking lane change proposal, i.e., a proposal of changing lanes to move the own vehicle **100** to a next lane to overtake the preceding vehicle **200**. In the present embodiment, the proposal process of the overtaking lane change is a process of displaying the overtaking lane change proposal on the display **31** and/or outputting the overtaking lane change

proposal from the speaker **32**.

(25) Next, the vehicle control apparatus **10** proceeds with the process to a step **S215** to determine whether or not the driver performs an approval operation. In the present embodiment, the approval operation at the step **S215** is an accelerator operation or an acceleration operation, i.e., an operation applied to the accelerator pedal **41** by the driver or a set vehicle travelling speed increase operation or an acceleration operation, i.e., an operation applied to the set vehicle travelling speed increase button **43**.

(26) When the vehicle control apparatus **10** determines “Yes” at the step **S215**, the vehicle control apparatus **10** proceeds with the process to a step **S220** to terminate executing the proposal process of the overtaking lane change, and then proceeds with the process to a step **S225** to execute the overtaking lane change control. The overtaking lane change control is a lane change control of controlling the steering force applied to the own vehicle **100** to change lanes to move the own vehicle **100** to the next lane. Next, the vehicle control apparatus **10** proceeds with the process to a step **S295** to terminate executing this routine once.

(27) That is, when the acceleration operation is performed by the driver when the vehicle control apparatus **10** provides the driver with a proposal of changing lanes to move the own vehicle to the next lane to overtake the preceding vehicle **200**, the vehicle control apparatus **10** determines that the driver approves of the proposal of changing lanes to move the own vehicle **100** to the next lane and executes the overtaking lane change control.

(28) On the other hand, when the vehicle control apparatus **10** determines “No” at the step **S215**, the vehicle control apparatus **10** proceeds with the process to a step **S230** to determine whether or not a rejection operation is performed by the driver. In the present embodiment, the rejection operation at the step **S230** is a brake operation or a deceleration operation, i.e., an operation applied to the brake pedal **42** or a set vehicle travelling speed decrease operation or a deceleration operation, i.e., an operation applied to the set vehicle travelling speed decrease button **44** by the driver.

(29) When the vehicle control apparatus **10** determines “Yes” at the step **S230**, the vehicle control apparatus **10** proceeds with the process to a step **S235** to terminate executing the proposal process of the overtaking lane change, and then proceeds with the process to the step **S295** to terminate executing this routine once.

(30) That is, when the driver performs the deceleration operation when the vehicle control apparatus **10** provides the driver with the proposal of changing lanes to move the own vehicle **100** to the next lane, the vehicle control apparatus **10** determines that the driver rejects the proposal of changing lanes to move the own vehicle **100** to the next lane and does not execute the overtaking lane change control.

(31) On the other hand, when the vehicle control apparatus **10** determines “No” at the step **S230**, the vehicle control apparatus **10** proceeds with the process to the step **S295** to terminate executing the process of this routine once.

(32) It should be noted that the vehicle control apparatus **10** may be configured to determine that the driver rejects the overtaking lane change proposal and terminate executing the proposal process of the overtaking lane change when an overtaking proposal time **T1**, i.e., a time elapsed from starting executing the proposal process of the overtaking lane change at the step **S210** becomes equal to or greater than a predetermined time (or a predetermined overtaking proposal time **T1\_th**). Of course, in this case, the overtaking lane change control is not performed.

(33) In addition, when the vehicle control apparatus **10** determines “No” at the step **S205**, the vehicle control apparatus **10** proceeds with the process to a step **S240** to determine whether or not a return lane change condition **C2** is satisfied.

(34) As shown in FIG. **4**, the return lane change condition **C2** is a condition that the driver is presumed to desire to perform a return lane change control described later. More specifically, the return lane change condition **C2** is, for example, a condition that a space where the own vehicle

**100** can enter exists ahead of the preceding vehicle **200** after the own vehicle **100** overtakes the preceding vehicle **200**. The vehicle control apparatus **10** determines whether or not a space where the own vehicle **100** can enter exists ahead of the preceding vehicle **200** enters based on the surrounding detection information IF. Further, when the overtaking lane change condition C1 includes a condition that the lane keeping control is being executed, the return lane change condition C2 also includes a condition that the lane keeping control is being executed.

(35) When the vehicle control apparatus **10** determines “Yes” at the step **S240**, the vehicle control apparatus **10** proceeds with the process to a step **S245** to execute a proposal process of a return lane change. The proposal process of the return lane change is a return lane change proposal, i.e., a process of providing the driver with a proposal of enter the own vehicle **100** into the space ahead of the preceding vehicle **200**. In the present embodiment, the proposal process of the return lane change is a process of displaying the return lane change proposal on the display **31** and/or outputting the return lane change proposal from the speaker **32**.

(36) Next, the vehicle control apparatus **10** proceeds with the process to a step **S250** to determine whether or not the driver performs an approval operation. In the present embodiment, the approval operation at the step **S250** is the braking operation (i.e., the deceleration operation) or the set vehicle travelling speed decrease operation (i.e., the deceleration operation).

(37) When the vehicle control apparatus **10** determines “Yes” at the step **S250**, the vehicle control apparatus **10** proceeds with the process to a step **S255** to terminate executing the proposal process of the return lane change, and then proceeds with the process to a step **S260** to execute the return lane change control. The return lane change control is a lane change control of controlling the steering force applied to the own vehicle **100** to change lanes to return the own vehicle **100** to an original lane by entering the own vehicle **100** into the space ahead of the preceding vehicle **200**. Next, the vehicle control apparatus **10** proceeds with the process to the step **S295** to terminate executing this routine once.

(38) That is, when the deceleration operation is performed by the driver when the vehicle control apparatus **10** provides the driver with a proposal of changing lanes to return the own vehicle **100** to the original lane after the own vehicle **100** moves to the next lane and overtakes the preceding vehicle **200**, the vehicle control apparatus **10** determines that the driver approve of the proposal of changing lanes to return the own vehicle **100** to the original lane and execute the return lane change control.

(39) On the other hand, when the vehicle control apparatus **10** determines “No” at the step **S250**, the vehicle control apparatus **10** proceeds with the process to a step **S265** to determine whether or not a rejection operation is performed by the driver. In the present embodiment, the rejection operation at the step **S265** is the accelerator operation (i.e., the acceleration operation) or the set vehicle travelling speed increase operation (i.e., the acceleration operation) by the driver.

(40) When the vehicle control apparatus **10** determines “Yes” at the step **S265**, the vehicle control apparatus **10** proceeds with the process to a step **S270** to terminate the proposal process of the return lane change, and then proceeds with the process to the step **S295** to terminate executing this routine once.

(41) That is, when the acceleration operation is performed by the driver when the vehicle control apparatus **10** provides the driver with the proposal of changing lanes to return the own vehicle **100** to the original lane, the vehicle control device determines that the driver rejects the proposal of changing lanes to return the own vehicle **100** to the original lane and does not execute the return lane change control.

(42) On the other hand, when the vehicle control apparatus **10** determines “No” at the step **S270**, the vehicle control apparatus **10** proceeds with the process to the step **S295** to terminate executing this routine once.

(43) The vehicle control apparatus **10** may be configured to determine that the driver rejects the return lane change proposal and terminate executing the proposal process of the return lane change



when a return proposal time T2, i.e., a time elapsed from starting executing the proposal process of the return lane change at the step S245 becomes equal to or greater than a predetermined time (i.e., a predetermined return proposal time T2\_th). Of course, in this case, the return lane change control is not executed. Further, the predetermined return proposal time T2\_th may be set to a time equal to or different from the predetermined overtaking proposal time T1\_th.

(44) It should be noted that the present invention is not limited to the above-described embodiments, and various modifications can be adopted within the scope of the present invention.

## Claims

1. A vehicle control apparatus comprising an electronic control unit which executes a lane change control of providing a driver of an own vehicle with a proposal of changing lanes and automatically change lanes when the driver approves of the proposal, wherein the electronic control unit is configured to: when the driver performs an acceleration operation when the electronic control unit provides the driver with a proposal of changing lanes to move the own vehicle to a next lane for overtaking a preceding vehicle, determine that the driver approves of the proposal of changing lanes to move the own vehicle to the next lane; when the driver performs a deceleration operation when the electronic control unit provides the driver with a proposal of changing lanes to return the own vehicle to an original lane after the own vehicle moves to the next lane and overtakes the preceding vehicle, determine that the driver approves of the proposal of changing lanes to return the own vehicle to the original lane.
  2. The vehicle control apparatus according to claim 1, wherein the electronic control unit is configured to: when the driver performs the deceleration operation when the electronic control unit provides the driver with the proposal of changing lanes to move the own vehicle to the next lane, determine that the driver rejects the proposal of changing lanes to move the own vehicle to the next lane; when the driver performs the acceleration operation when the electronic control unit provides the driver with the proposal of changing lanes to return the own vehicle to the original lane, determine that the driver rejects the proposal of changing lanes to return the own vehicle to the original lane.
  3. The vehicle control apparatus according to claim 1, wherein the electronic control unit is configured to provide the driver with the proposal of changing lanes to move the own vehicle to the next lane when a distance between the own vehicle and the preceding vehicle becomes equal to or smaller than a predetermined distance, and a travelling speed of the own vehicle is greater than a travelling speed of the preceding vehicle.
  4. The vehicle control apparatus according to claim 1, wherein the electronic control unit is configured to provide the driver with the proposal of changing lanes to return the own vehicle to the original lane when a space where the own vehicle can enter exists ahead of the preceding vehicle after the vehicle moves to the next lane and overtakes the preceding vehicle.
-